

**Guide for Estimating Subgrade Soil Strengths
(Fine-Grained Soils)**

Estimated Consistency by:		Tested by:	Correlates to:		
Feel	Visual	Standard Penetration Test (blows/foot)	Shear Strength, Cu		R Value AASHTO
			(psi)	(tsf)	(%)
Very Soft	Man Standing Sinks more than 3 inches	<2	<1.7	<0.125	<0.4
Soft	Man Walking Sinks about 2 to 3 inches	2 - 4	1.7 - 3.5	0.125 - 0.25	0.4 - 0.8
Medium	Man Walking Sinks About 1 inch	4 - 8	3.5 - 7	0.25 - 0.50	0.8 - 1.6
Stiff	Pickup Truck Ruts about 1/2 - 1 inch	8 - 15	7 - 14	0.50 - 1.0	1.6 - 3.2
Very Stiff	Loaded Dump Truck Ruts 1 - 3 inches	15 - 30	14 - 28	1.0 - 2.0	3.2 - 6.4
Hard	Insignificant Ruts from Loaed Dump Truck	>30	>28	>2.0	>6.4

After: Portland Cement Assocation, E I. DuPort Literature; McCarthy, David F., "Essentials of Soil Mechanics and Foundations," 1977: AASHTO, "1993 Guide for Design of Pavement Structures", Van Till et. al. NCHRP 128.